1. Clinical Question & Core Hypothesis

Clinical Question

Why do organized local immune structures (TLS/TIL) in MIBC sometimes fail to predict benefit from immune checkpoint blockade—i.e., under what tumor-intrinsic conditions does local immune organization translate into a systemic, therapy-responsive anti-tumor immune state? 2) Tier 1 paths used: I built on the bTMB and MSH6/TMB core paths because they form a mechanistic triad linking DNA-repair status (MSH6) to neoantigen generation (TMB/bTMB) and thereby to the effectiveness of pre-existing immune infiltrates. bTMB was prioritized as an anchor alongside MSH6 to emphasize a clinically deployable, non-invasive readout that captures systemic mutational heterogeneity relevant to therapy response. 3) Tier 2 evidence that supports the claim: PD-L1 and S100A9 link innate/inflammatory regulation to checkpoint expression and may modulate whether lymphoid aggregates are functionally suppressive or activating; MSI/dMMR provides a mechanistic precedent for MMR-driven neoantigenicity; AI/accessibility arguments justify using bTMB as a scalable assay to triage patients. 4) Endpoint choice rationale: I selected therapy_response (immune checkpoint therapy response) because the core mechanistic question connects neoantigen load and MMR status to immunotherapy sensitivity, and bTMB/msr biomarkers are most actionable in the context of treatment selection. 5) Deliberate exclusions: I excluded exploratory metabolic and circulating metabolic marker paths (ALP/HDL-C/TG/SHBG/IGF-1), stool methylation markers (SDC2/TFPI2), and peripheral prognostic signals (pZVR) from the core hypothesis because they are indirect, cross-tissue, or insufficiently mechanistic for the TLS–TMB–MMR interaction I focus on. I also relegated PD-L1, S100A9, TP53, MSI/dMMR and GARS1 to secondary modifiers rather than anchors to keep the hypothesis sharply mechanistic and testable?

Working Hypothesis

Primary refined hypothesis: In urothelial bladder cancer (MIBC), the presence of TLS-like lymphoid aggregates predicts responsiveness to immune checkpoint blockade only when the tumor exhibits high tumor-intrinsic neoantigen generation driven by MSH6 deficiency and elevated systemic neoantigen load as captured by high bTMB. Specifically, TLS-high, MSH6-deficient, and bTMB-high tumors will show superior objective response rates and longer progression-free survival to anti-PD-1/PD-L1 therapy compared with TLS-high tumors that are MSH6-proficient or have low bTMB, and TLS-high status alone will not reliably predict response in the absence of high neoantigen burden. This core relationship is modulated by secondary factors such as PD-L1 expression, S100A9+ myeloid cells, MSI/dMMR status, TP53 alterations, and GARS1, which can amplify or dampen the TLS-driven immunogenic potential but are not required for the fundamental TLS–neoantigen–ICI axis. Mechanistically, TLS provide localized antigen presentation and T cell priming, but productive systemic anti-tumor immunity under checkpoint blockade requires a sufficient pool of neoantigens to drive clonal expansion of neoantigen-specific T cells; MSH6 deficiency increases mutational load and neoantigen diversity, enabling TLS-driven priming to translate into systemic, therapy-responsive immunity. bTMB serves as a clinically actionable surrogate for systemic neoantigen load that, when integrated with TLS status and MSH6, yields improved prediction of ICI response beyond either biomarker alone. The hypothesis generates testable predictions: (i) TLS-high/MSH6-deficient/bTMB-high tumors will have higher response rates and longer PFS than TLS-high/MSH6-proficient/bTMB-low tumors; (ii) TLS-high/MSH6-deficient/bTMB-high tumors will show enriched cytotoxic T cell effector signatures and higher TCR clonality expansion post-therapy; (iii) TLS-high with low bTMB or intact MSH6 will not consistently predict response, even if TLS are present.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, bTMB
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6/TMB core paths because they form a mechanistic triad linking DNA-repair status (MSH6) to neoantigen generation (TMB/bTMB) and thereby to the effectiveness of pre-existing immune infiltrates.

bTMB: bTMB and MSH6/TMB core paths because they form a mechanistic triad linking DNA-repair status (MSH6) to neoantigen generation (TMB/bTMB) and thereby to the effectiveness of pre-existing immune infiltrates.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PD-L1	Supporting modifier from knowledge graph
Secondary	S100A9	Supporting modifier from knowledge graph
Secondary	MSI/dMMR	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , MSH6 sequencing, targeted or whole-exome sequencing for bTMB, cfDNA-based bTMB if tissue is unavailable, RNA-seq/spatial transcriptomics for TLS activity, TCR repertoire analysis, and preclinical models with MSH6 manipulation.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 3: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- S100A9: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.